Chapter 13

Problems 4

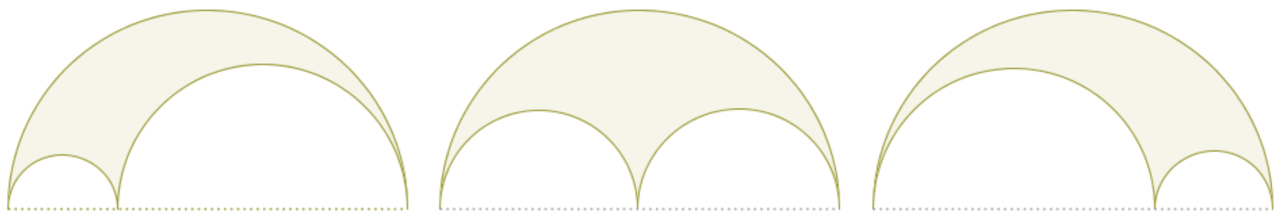
In this, also short, chapter we will, again, solve only one problem, a problem that deals with the so-called **arbelos** or a planar region that is trapped between three bounding semicircles. However, the length of this chapter is deceiving because just this one problem alone also punches above its weight. In order to see that, we highly recommend that our readers try solve this problem in any which other way, a way that does not rely on the machinery of inversive geometry.

While the upcoming problem may be not as challenging as the Chain Of Pappus problem that we slayed in the previous chapter, it is still difficult enough and it is that difficulty of generating the required proofs that rely on "traditional" methods is what brings to light, one more time, the awesome power of the "Transform And Conquer" problem-solving approach in general and the inversive geometry in particular.

Transform. Solve. Reverse-transform.

Before we deep-dive into the underlying mathematics of the upcoming problem, let us agree on an informal definition of what **an arbelos** is.

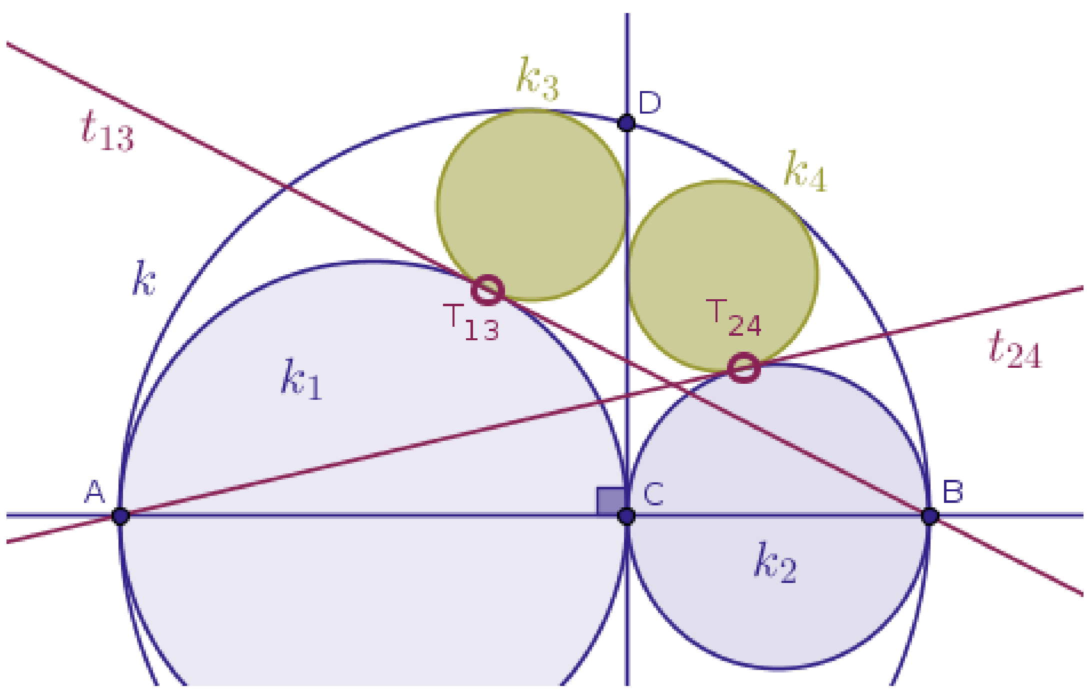
An arbelos is a (green) curvilinear planar region that is trapped, logically, below a parent semicircle and, logically, above the two semicircles that are constructed on the diameter of the parent semicircle (Figure 13.1):



Due to the symmetry of the above construction it is evident that the point where the two smaller semicircles touch each other (externally) can be located, pretty much, anywhere on the generating diameter of the parent (larger) semicircle.

So far we have explored the application of various inversions with respect to a circle with **positive** power to problem-solving. In this chapter, as a closing sample of our introductory text, we shall investigate a problem that puts an inversion with respect to a circle with, specifically, **negative** power front and center.

Problem 19 (an Arbelos Encounter). Among the three given distinct collinear points, A, B, C, the point C is located between the points A and B (Figure 13.2):



On the line segments AB, CA and CB as diameters, the circles k, k_1 and k_2, correspondingly, are constructed in such a way that the circles k_1 and k_2 touch each other externally at the point C, while touching the circle k at the points A and B internally, correspondingly.

A straight line through the point C perpendicular to the straight line AB intersects the circle k at the point D.

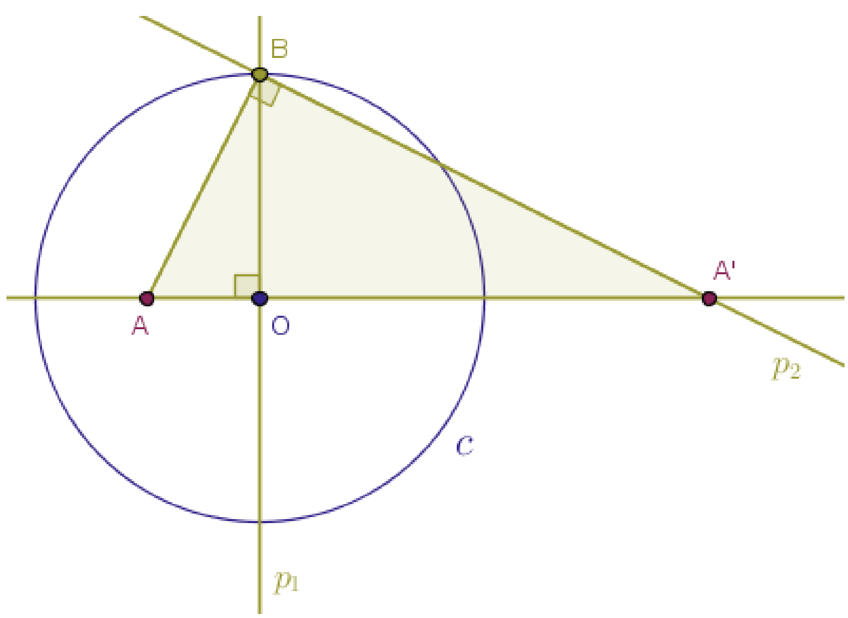
Into thus formed curvilinear triangles ADC and BDC two more circles, k_3 and k_4, are inscribed in such a way that the circle k_3 touches the circle k_1 at the point T_{13} externally and the circle k_4 touches the circle k_2 at the point T_{24} also externally as shown in Figure 13.2 above.

Prove that:

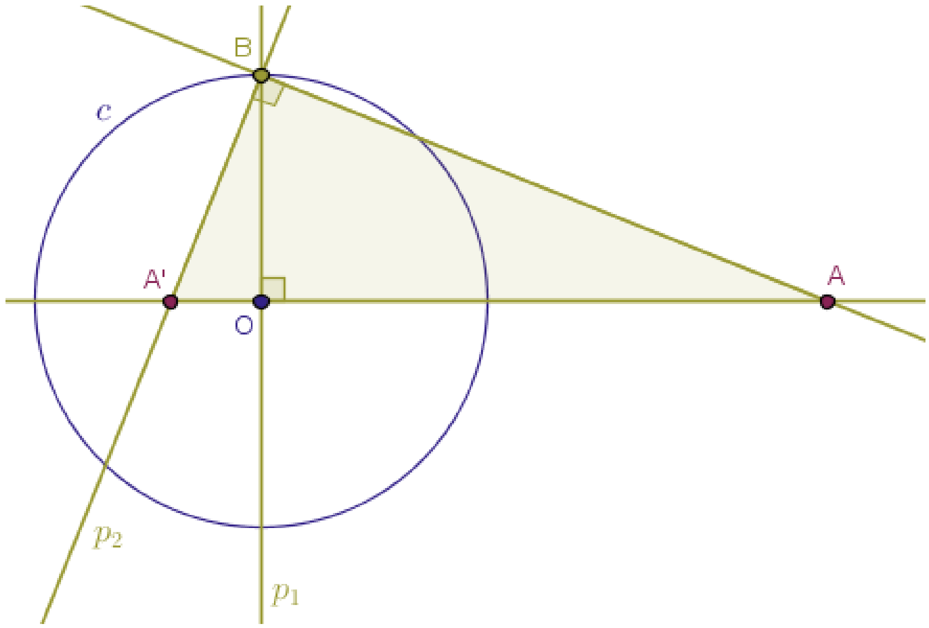
- **Part 1:** the lengths of the radii of the circles k_3 and k_4 are equal
- **Part 2:** the internal tangent t_{13} common to the circles k_1 and k_3 at the point T_{13} passes through the point B, while the internal tangent t_{24} common to the circles k_2 and k_4 at the point T_{24} passes through the point A

Proofs. Part 1. We first prove that the circles k_3 and k_4 are, in fact, equal in the sense that their radii are equal in length. To that end we recall our Problem 2 and Problem 4 from the Chapter 3, Points, Part 1/2, where we learned how to construct an inverse image of a given point with respect to a certain circle with, specifically, negative (!) power.

That is, in the Problem 2 we obtained an inverse image of a point, A, with respect to a circle, c, with negative power when the given point was located **in**side of the circle of inversion (Figure 3.7):



and conversely. In the Problem 4 we obtained an inverse image of a point, A, with respect to a circle, c, with negative power when the given point was located **out**side of the circle of inversion (Figure 3.12):



While solving the above two problems, we learned that regardless of where the original point is located, inside or outside of the circle of inversion, the interior angle of the triangle ABA' at the helper vertex B was always right.

How did we construct the said auxiliary point B?

We constructed the helper point B as a by-product of the intersection of the circle of inversion, c, with the straight line that was **perpendicular** to the straight line OA at the center of inversion, the point O.

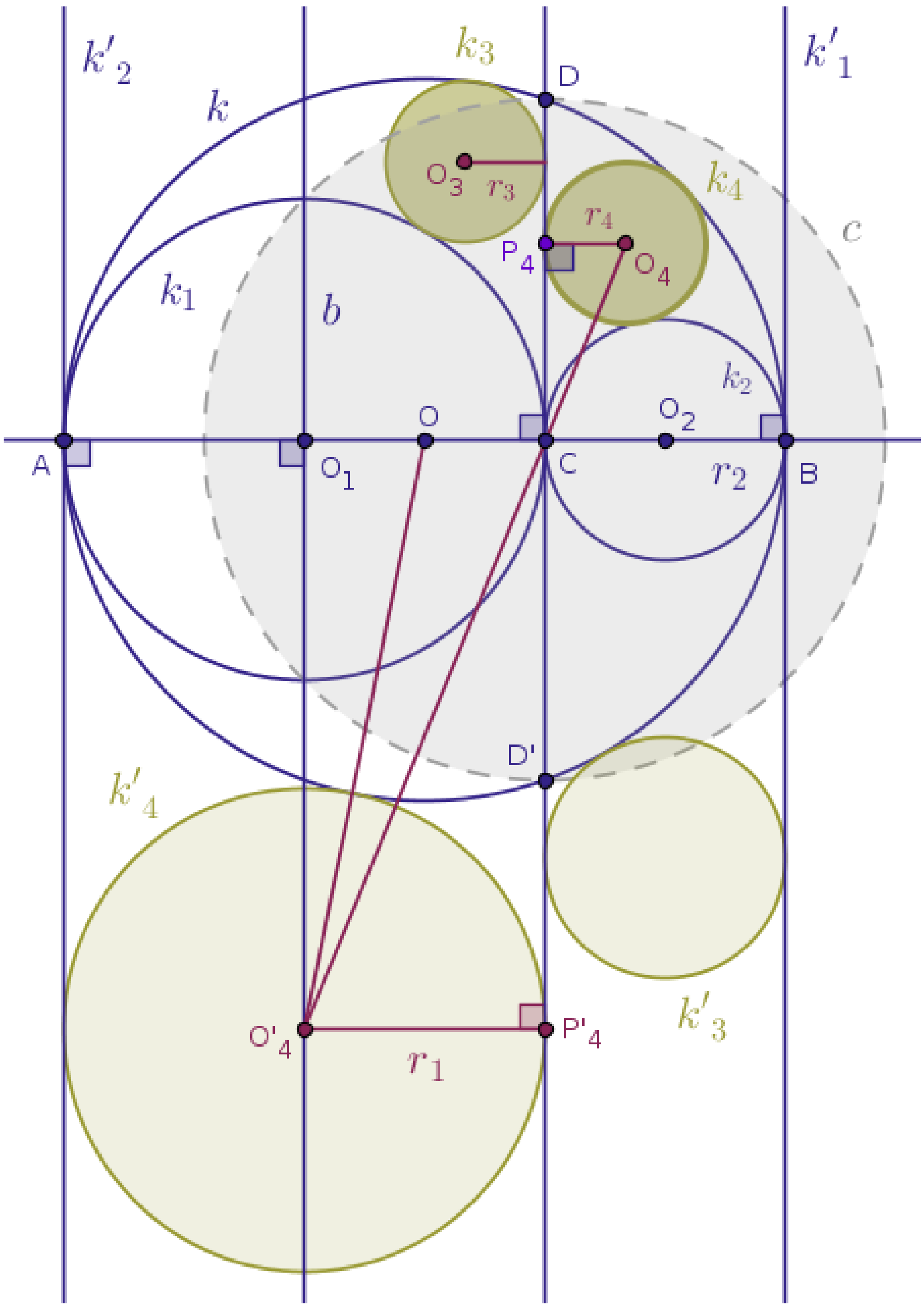
If we juxtapose the constructions shown in the above figures against the construction that renders the input for the given problem shown in Figure 13.2 above then we will have no choice but to observe that since the angle ADB in Figure 13.2 subtends the diameter AB of the input circle k, then, by Euclid's "Elements" Book 3 Proposition 31, the said angle ADB must be right.

Hence, working in reverse order, we may very well **demand** that the given points A and B are, in fact, the inverse images of each other.

Inverse images of each other with respect to what circle and what power?

Well, we are given that the straight line CD is perpendicular to the straight line AB. Therefore, our mystery circle of inversion, c, reveals itself as a circle centered at the point C with the radius CD: $c(C, |CD|)$.

Moreover, the power of inversion under which the points A and B are the reflections of each other in the circle c is **negative**. Thus, as we have discussed this idea already, we now want to enumerate the inverse images of all the input objects of interest, as they are reflected in the gray circle $c(C, |CD|)$ with negative power (Figure 13.3):



Under the said inversion, where we take it that the circles $c(C, |CD|)$ and k intersect at the points D and D':

- the straight lines AB and CD will be taken into themselves (Theorem 9)
- the generating circle, k, will also be taken into itself (Theorem 42)
- the inscribed circle k_1 will be taken into a straight line k_1' that will be **perpendicular** to the straight line AB and that will **touch** the generating circle, k, at the point B (Theorems 16 and 38)
- the inscribed circle k_2 will be taken into a straight line k_2' that will be perpendicular to the straight line AB and that will touch the generating circle, k, at the point A (Theorems 16 and 38)
- the inscribed circle k_3 will be taken into a circle k_3' that will touch:
    - the straight line CD
    - the generating circle, k, and
    - the straight line k_1' (Theorems 24 and 38)
- the inscribed circle k_4 will be taken into a circle k_4' that will touch:
    - the straight line CD
    - the circle k and
    - the straight line k_2' (Theorems 24 and 38)

Further, let the points O, O_1, O_2, O_3, O_4 and O_4' be the centers of the circles k, k_1, k_2, k_3, k_4 and k_4' correspondingly.

Moreover, let b be the straight line passing through the point O_1 perpendicular to the straight line AB.

Let the straight line passing through the point O_4 perpendicular to b intersect the straight line CD at the point P_4. Likewise, let the straight line passing through the point O_4' perpendicular to b intersect the straight line CD at the point P_4'.

Then, clearly, for the lengths of the radii $r_{1,2}$ of the circles $k_{1,2}$, we have:

$$|O_1A| = |O_1C| = |O_4'P_4'| = r_1$$

and:

$$|O_2C| = |O_2B| = r_2$$

where $r_{1,2}$ are the real constant or fixed numbers.

If r is the length of the radius of the generating circle k then:

$$|OA| = |OB| = r = \frac{2r_1 + 2r_2}{2} = r_1 + r_2$$

which is to say that:

$$r_1 + r_2 = r \tag{1}$$

The idea now is to express the lengths of the radii $r_{3,4}$ of the circles $k_{3,4}$ via the magnitudes of $r_{1,2}$ and show that it is the case that $r_3 \equiv r_4$.

We will work with the inscribed circle k_4 together, while the reader is encouraged to generate her/his own analysis for the inscribed circle k_3 in the similar fashion.

To that end, consider the circles k_4 and k'_4 that have the straight line CD as their common interior tangent. Since in the right triangles CP_4O_4 and $CP'_4O'_4$ the interior angles at the vertex C are vertical, the said triangles are similar by **AAA**.

In such triangles, by Euclid's "Elements" Book 6 Proposition 4, the ratios of the lengths of the sides about the equal angles, where the corresponding sides are opposite the equal angles, are the same:

$$\frac{|O_4P_4|}{|CP_4|} = \frac{|O'_4P'_4|}{|CP'_4|}$$

which is to say that:

$$\frac{r_4}{|CP_4|} = \frac{r_1}{|CP'_4|}$$

from where it follows that:

$$r_4 = r_1 \cdot \frac{|CP_4|}{|CP'_4|} \tag{2}$$

In order to recover the lengths $|CP_4|$ and $|CP'_4|$ in terms of $r_{1,2}$, we observe that the points P_4 and P'_4 are the inverse images of each other with respect to the circle c with negative power.

Hence, by the Condition 3 of Definition 1 and our definition of an inversion with respect to a circle with negative power, for these two points we have:

$$|CP_4| \cdot |CP'_4| = |CD|^2$$

from where:

$$|CP_4| = \frac{|CD|^2}{|CP'_4|} \tag{3}$$

Putting the RHS of (**3**) into the RHS of (**2**), we find:

$$r_4 = r_1 \cdot \frac{|CD|^2}{|CP_4'|^2} \tag{4}$$

By exactly the same reasoning, since the points A and B are the inverse images of each with respect to the circle c with negative power, we find that:

$$|CA| \cdot |CB| = 2r_1 \cdot 2r_2 = |CD|^2 \tag{5}$$

which is to say officially that:

$$|CD| = 2\sqrt{r_1 r_2} \tag{6}$$

Putting the RHS of (**6**) back into the RHS of (**4**), we arrive at:

$$r_4 = \frac{4r_1^2 r_2}{|CP_4'|^2} \tag{7}$$

In order to recover the magnitude of the square $|CP_4'|^2$, we apply the Pythagorean theorem to the right triangle OO_1O_4':

$$|CP_4'|^2 = |O_1O_4'|^2 = |OO_4'|^2 - |OO_1|^2 \tag{8}$$

But, clearly, since the circles k and k_4' touch, we have for the distance $|OO_4'|$:

$$|OO_4'| = r + r_1$$

which, with the relation in (**1**), implies that:

$$|OO_4'| = r + r_1 = r_1 + r_2 + r_1 = 2r_1 + r_2 \tag{9}$$

Correspondingly, for the distance $|OO_1|$ we have:

$$|OO_1| = |OA| - |O_1A| = r - r_1$$

which, with the relation in (1), implies that:

$$|OO_1| = r_1 + r_2 - r_1 = r_2 \tag{10}$$

Putting the RHS images of (**9**) and (**10**) back into (**8**), we find:

$$|CP_4'|^2 = (2r_1 + r_2)^2 - r_2^2$$

which is to say that:

$$|CP_4'|^2 = 4r_1\left(r_1 + r_2\right) \tag{11}$$

Putting the RHS of (**11**) into the RHS of (**7**), we see that:

$$r_4 = \frac{r_1 r_2}{r_1 + r_2} \tag{12}$$

By exactly same reasoning, but this time starting with the circles k_3 and k_3' that still have the straight line CD as their common interior tangent, we show that:

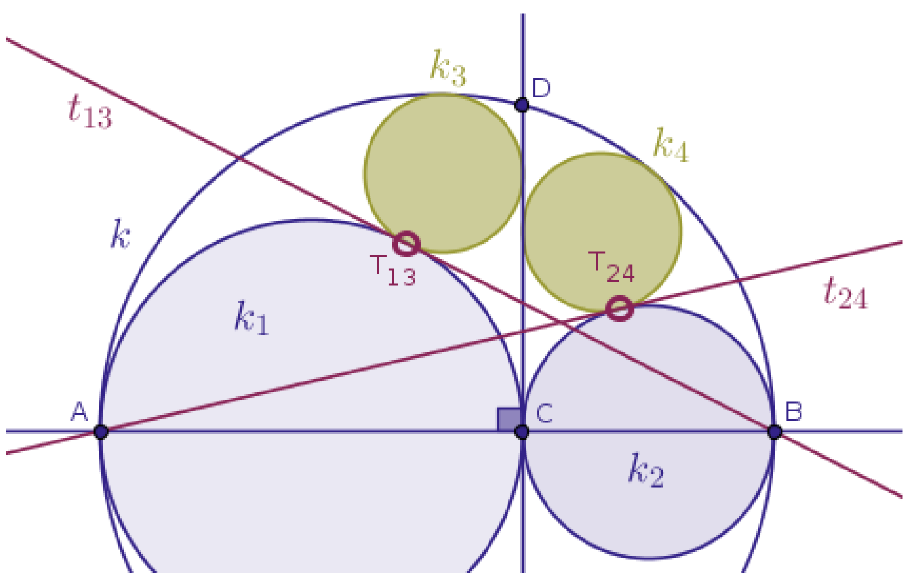
$$r_3 = \frac{r_1 r_2}{r_1 + r_2} \tag{13}$$

Therefore, it is the case that:

$$r_3 \equiv r_4$$

implying that the lengths of the radii $r_{3,4}$ of the inscribed circles $k_{3,4}$ are, in fact, equal, which is what was required to prove. □

Part 2. In the second part of our proofs we want to show that the straight line t_{13} that plays the role of a common internal tangent to the circles k_1 and k_3 at the point T_{13} must necessarily pass through the point B, while the straight line t_{24} that plays the role or a common internal tangent to the circles k_2 and k_4 at the point T_{24} must necessarily pass through the point A (Figure 13.2):

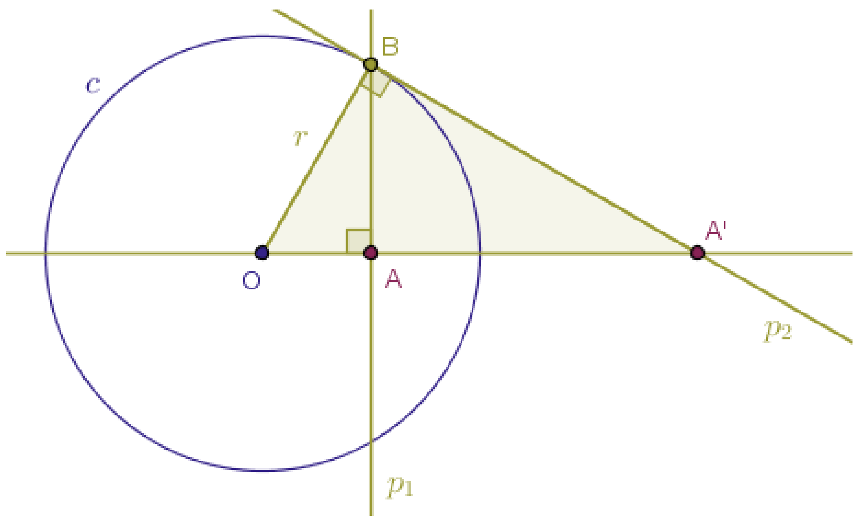


To that end, recall that in the first part of our proof we demanded that the points A and B were the inverse images of each with respect to a mystery circle, c, with negative power.

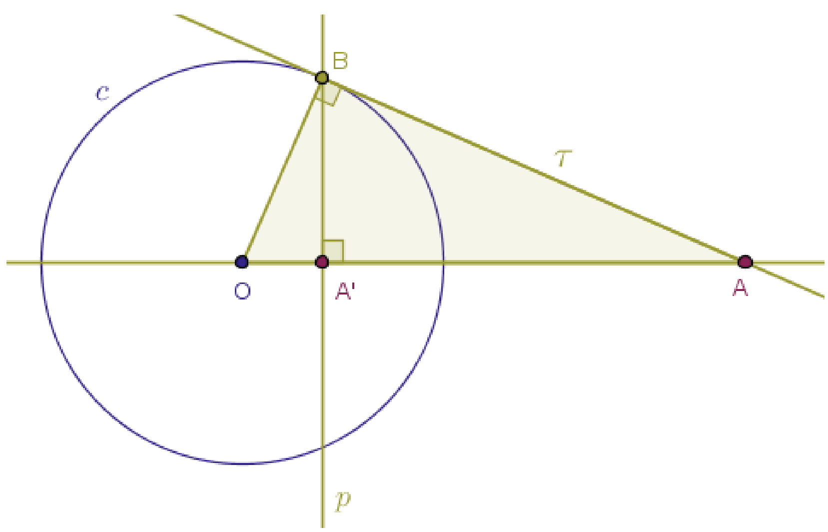
Now, however, we switch gears a little bit and demand that the points A and C be the inverse images of each other with respect to a mystery circle, c, with positive (!) power.

Note that even though we are shuffling the center and the power of inversion around, the angle ADB or BDA still remains right!

Therefore, from the relevant mechanics that we have studied in the Problem 1 (Figure 3.5):



and in Problem 3 (Figure 3.10):

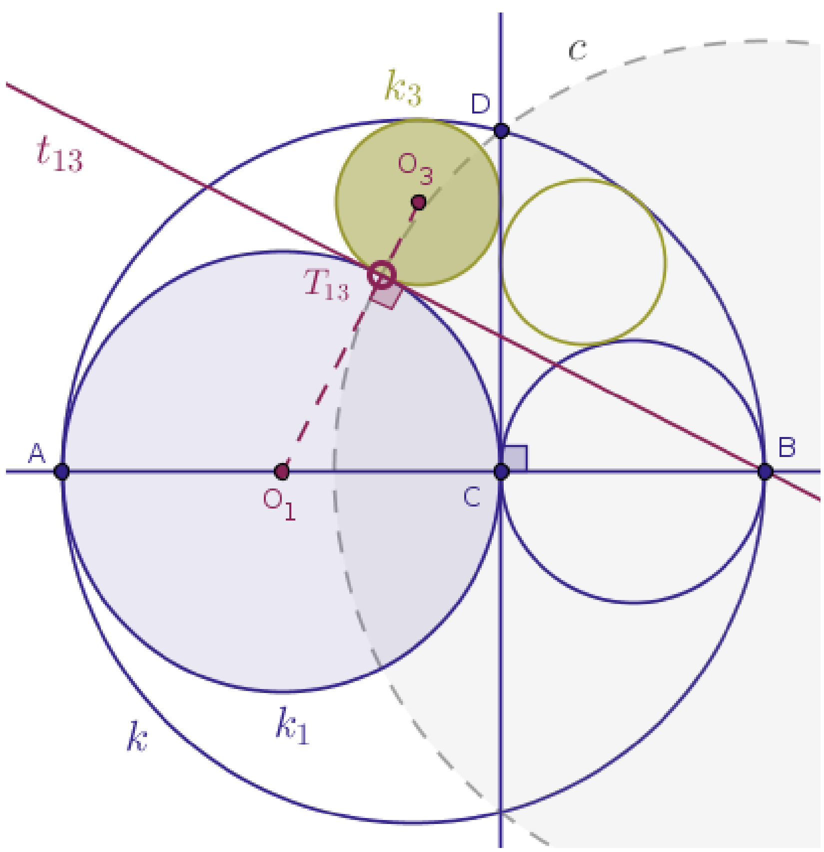


we conclude that the given points A and C will be the inverse images of each other under the inversion with respect to the circle c centered at the point B with the radius BD: $c(B, |BD|)$.

We now study the amazing tango of the relevant inverse objects. Under the inversion with respect to the circle $c(B, |BD|)$ with positive power:

- the straight line AB will be taken into itself (Theorem 9)
- the straight line CD will be taken into the circle k (Theorem 10)
- the circle k will be taken into the straight line CD (Theorem 15)
- the circle k_1 will be taken into itself (Theorems 39 and 41)
- hence, the curvilinear triangle ADC will be taken into itself
- (now, how neat is that? watch what happens next)
- the circle k_3 that touches the objects k, k_1 and CD and is inscribed into the curvilinear triangle ADC will be taken into itself! (Theorems 22, 37, 38)

We capture the above analysis in Figure 13.4:



Lastly, in the proof of the Theorem 39 we saw that since the circle k_3 remains invariant under the inversion with respect to the circle c with positive power, it follows that these two circles, k_3 and c, must be orthogonal.

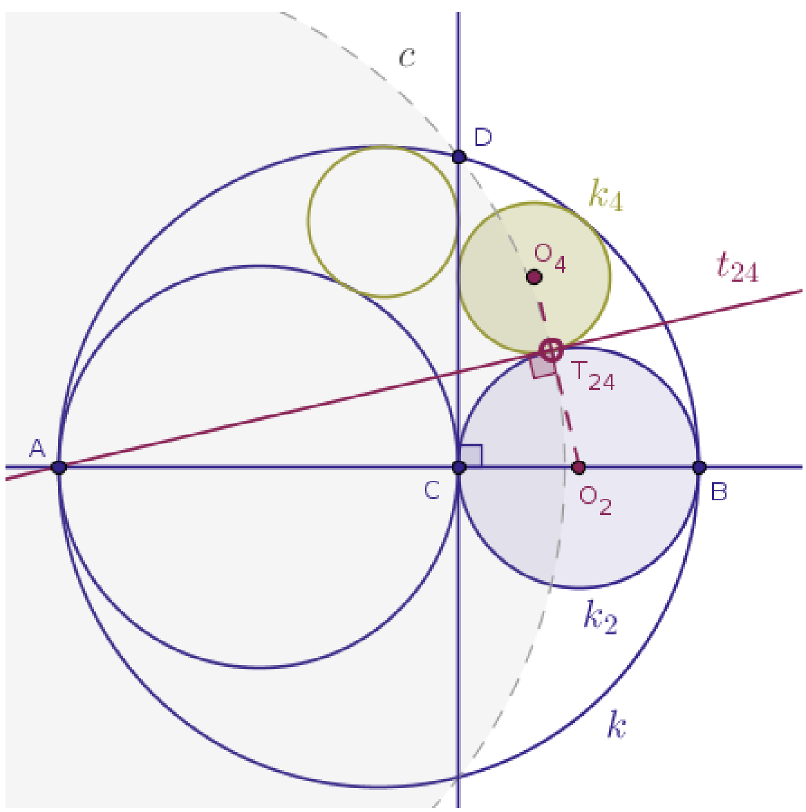
Likewise, the circle k_1 is orthogonal to the circle c. Why. Under the chosen inversion, point-wise the circle k is taken into the straight line CD and, conversely, the straight line CD is taken into the circle k. But figure-wise these two geometric objects still remain what and where they are. Recall that we have seen the similar phenomenon in the Theorem 9.

It follows, then, that under the said inversion, the single point of tangency T_{13} of the circles k_1 and k_3 will be taken into itself, which is to say that the point T_{13} will be fixed.

But the circle c intersects the semi-circle $AT_{13}C$ at a single, fixed, point. Hence, the point T_{13} belongs to three circles at once: the circles c, k_1 and k_3. Of the said three circles, two circles, k_1 and k_3, touch externally at the point T_{13} by hypothesis and are orthogonal to the third circle, c.

Therefore, the straight line t_{13} that plays the role of a common internal tangent to the circles k_1 and k_3 at the point T_{13} must necessarily pass through the point B by the definition of orthogonal circles. □

The reader is encouraged to generate her/his own proof of the fact that the straight line t_{24} that plays the role or a common internal tangent to the circles k_2 and k_4 at the point T_{24} must necessarily pass through the point A. As a hint, you would want to consider the inversion with respect to the circle $c(A, |AD|)$ with positive power and analyze what will happen to the input objects of interest under that inversion (Figure 13.5):



At his point, officially ending the introductory part of this text, we would like to thank our readers for staying with us, wish you the best of luck with your studies and we hope to see you around as we have more interesting material to cover. □